Boltzmann Transport from Observer-Patch Consistency: a Theorem-Conditional Reduction

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Abstract

This note shows that standard general-relativistic photon–baryon Boltzmann transport follows on the Observer-Patch Holography (OPH) branch when its geometry, electromagnetic action, physical photon, electron mass, and recombination history are supplied. It locates the repair-exchange and collar-stress interfaces where an OPH-specific effect could enter. A physical cosmic microwave background (CMB) prediction requires finite source receipts, transfer functions, and likelihood closure.

1 Claim boundary

Everything below is conditional on the branch theorems it cites. The *transport form* used by the diagnostic calculation with the Code for Anisotropies in the Microwave Background (CAMB) is a theorem-conditional reduction with declared imports. It does not certify the initial spectrum (amplitude or tilt), the recombination history, the absolute Thomson normalization, or any likelihood statement. Those require a finite primordial source, a physical radial lift, active-kernel and clock receipts, and an immutable likelihood contract. No statement in this note reads Planck data.

2 Branch inputs

Assumption 2.1 (Einstein branch)**.** *The observer-facing normal form carries a Lorentzian metric g_{ab} satisfying $G_{ab} + \Lambda g_{ab} = 8\pi G_{\mathrm{geom}}\langle T_{ab}\rangle$, with $G_{\mathrm{geom}} = \ell_\star^2$ from the edge-entropy/area-law theorem and $\Lambda\ell_\star^2 = 3\pi/N_\star$ from capacity closure. $N_\star$ is the capacity-identity estimate read from measured Λ through a declared capacity-to-scale bridge; Λ is an input to this assumption, not an OPH output. A physical interpretation of $N_\star$ beyond that estimate requires the readback map F and its contraction certificate. The cited branch theorem supplies the other geometry and capacity premises.*

Assumption 2.2 (Electromagnetic action and photon pole)**.** *The realized branch carries the Standard Model quotient and the Maxwell kinetic action with a* U(1) *vertex of strength α. This note uses $\alpha = \alpha_{\mathrm{root}}$, the exact root of a declared source map, conditional on same-scheme transport, target-independent selection of that map, and a typed bridge showing that its two coordinates read the same physical quantity. In addition, this transport note assumes a positive-energy physical QED quantization with two transverse states and a positive-residue isolated pole at $k^2 = 0$ on the ordinary Lorentz-vacuum, unbroken electromagnetic phase. This extra photon-pole receipt is not implied by the reconstructed abstract group. Charged-lepton* ratios *are branch outputs; the absolute electron* *mass m_e is empirically anchored. The inputs are the cited electroweak branch theorem, the explicit quantum-pole assumption just stated, and the declared import m_e.*

Assumption 2.3 (Cosmological background)**.** *The homogeneous limit of the branch is spatially flat FRW, $\mathrm{d}s^2 = a^2(\tau)\,[-\mathrm{d}\tau^2 + \delta_{ij}\mathrm{d}x^i\mathrm{d}x^j]$, with Friedmann dynamics from Assumption 2.1. Scalar perturbations are written in the conformal Newtonian gauge,*

$$\mathrm{d}s^2 = a^2(\tau)\left[-(1+2\psi)\,\mathrm{d}\tau^2 + (1-2\phi)\,\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j\right],$$

with conventions of Ma–Bertschinger (1995). Overdots are ∂_τ.

Assumption 2.4 (Recombination import)**.** *The free-electron history $n_e(\tau)$ (equivalently $x_e(\tau)$) is computed from standard atomic physics (Saha/Peebles-class) using imported atomic data. This is the same import class as the cesium-clock wall in the gravity stack: it contains no CMB-target information, so it is a permissible declared input under the anti-fitting rule, but it is an import, not a branch output.*

Assumption 2.5 (Conditional initial spectrum)**.** *Initial curvature perturbations $\zeta(\mathbf{k})$ are Gaussian with spectrum $\Delta^2_\zeta(k) = A_s\,(k/k_*)^{n_s-1}$. The Gaussian and statistically isotropic character is the maximum-entropy screen-covariance theorem. A finite tilt certificate requires a source generator, the weighted orientation-half identity, and a physical clock receipt. The radial theorem proves that one-shell inversion is nonunique and supplies physical source-dilation and cross-covariance tomography as conditional uniqueness routes. Finite source instantiation is not supplied here. This note consumes A_s and n_s as gated inputs and derives no initial spectrum.*

3 Conditional Refinement Liouville Transport

Definition 3.1 (One-photon record density)**.** *Conditional on the physical pole in Assumption 2.2, let $\mathcal{E}_\gamma$ be its positive-energy photon sector. Between record commits, a single excitation is a record continued across patch overlaps. Coarse-graining patch position and overlap momentum labels defines the phase-space record density $f(x^i, q, \hat{n}, \tau)$ on the forward mass shell, where q is comoving momentum magnitude and $\hat{n}$ the direction.*

Theorem 3.2 (Refinement Liouville)**.** *On a branch with the Maxwell action, quantum photon-pole receipt, and overlap Lorentz kinematics from Assumptions 2.2 and 2.1, let P_r be the quotient-lumpable free-flow transition kernel at step h_r. For every frozen smooth compactly supported test function ϕ, set*

$$L_r\phi = \frac{P_r\phi - \phi}{h_r}, \qquad \Gamma_r(\phi) = \frac{P_r(\phi^2) - (P_r\phi)^2}{h_r}.$$

The photon refinement convergence condition consists of compact containment/tightness, convergence of initial laws, uniform-on-compact generator convergence $L_r\phi \to X_H\phi$ to the null-geodesic Hamiltonian vector field, vanishing quadratic variation $\Gamma_r(\phi) \to 0$, uniqueness of the Hamiltonian ODE, refinement and schedule naturality, and vanishing boundary and phase-space-measure defects. Under this receipt, the between-commit record flow satisfies, in the refinement limit,

$$\frac{Df}{\mathrm{d}\tau} \;\equiv\; \frac{\partial f}{\partial \tau} + \frac{\mathrm{d}x^i}{\mathrm{d}\tau}\frac{\partial f}{\partial x^i} + \frac{\mathrm{d}q}{\mathrm{d}\tau}\frac{\partial f}{\partial q} + \frac{\mathrm{d}\hat{n}}{\mathrm{d}\tau}\cdot\frac{\partial f}{\partial \hat{n}} \;=\; C[f],$$

where the characteristics are the null geodesics of g_{ab} and $C[f]$ is supported on record commits only. In particular the collisionless flow preserves the Liouville measure.

Proof sketch under the convergence condition. For every test function, the discrete process minus the time integral of $L_r\phi$ is a martingale whose quadratic variation is controlled by $\Gamma_r(\phi)$. Tightness gives subsequential limits and vanishing quadratic variation makes every limiting martingale deterministic. Generator convergence says that each limit solves the martingale problem for X_H, equivalently the null-geodesic Hamiltonian ODE. Uniqueness identifies the whole limit, independently of the accepted repair schedule. Thus $\mathrm{d}x/\mathrm{d}\tau$, $\mathrm{d}q/\mathrm{d}\tau$, and $\mathrm{d}\hat{n}/\mathrm{d}\tau$ are the geodesic-flow coefficients. On the background of Assumption 2.3, $\mathrm{d}q/\mathrm{d}\tau = q[\dot{\phi} - \ldots]$ has the form used in §5.

Measure preservation. The identified limit is Hamiltonian flow on the mass shell, so it preserves the canonical Liouville measure. The finite measure-defect clause checks that the quotient measures converge to that canonical measure rather than to a different coarse volume form. Accepted repair confluence only removes schedule dependence of committed endpoints. It supplies neither an inverse between-commit dynamics nor a phase-space Jacobian. The orientation doubling of a finite label register is bookkeeping and does not produce either property.

Convergence premise. Photon-sector convergence of the finite patch flow to the displayed generator and zero-quadratic-variation limits is not supplied here. The abstract Lean layer proves the corresponding schedule-independence and refinement-naturality statement in its setting. Section 7 gives the corresponding simulator test. Theorem 3.2 is conditional on this convergence premise. □

Corollary 3.3 (Collision and diffusion alternatives)**.** *If the finite generator decomposes as $L_r = L_r^{\mathrm{free}} + C_r$ and $C_r\phi \to C\phi$, then the limiting density obeys*

$$\partial_\tau f + \{f, H\} = C^* f.$$

A nonzero limit of Γ_r produces a diffusion/Fokker–Planck term and does not satisfy the deterministic Liouville conclusion.

Remark 3.4. The transport model places the collision functional at record commits and uses the independently certified measure-preserving flow between them. Claims about entropy production therefore refer to the commit terms counted by $C[f]$. The accepted repair relation and the reversible equilibrium or between-commit dynamics are separate structures.

4 Record Commits and the Thomson Collision Functional

Theorem 4.1 (Scalar-intensity commit-collision functional)**.** *Let commits between photon-class and electron-class records occur with the soft-limit matrix element of the recovered* U(1) *vertex from Assumption 2.2. Then, to first order in perturbations and at low occupancy, the collision functional in Theorem 3.2 is the Thomson functional: in the electron rest frame,*

$$C[f](q,\hat{n}) \;=\; a\, n_e \sigma_T \left[\frac{3}{16\pi}\int \mathrm{d}\Omega' \,(1 + (\hat{n}\cdot\hat{n}')^2)\, f(q,\hat{n}') - f(q,\hat{n})\right], \qquad \sigma_T = \frac{8\pi}{3}\frac{\alpha^2}{m_e^2},$$

boosted to the frame where baryons move with velocity v_b. The source-derived reversible proposal kernel is assumed to obey detailed balance on the angular state space. Write/check orientation labels alone do not imply that property, and the accepted strict-descent repair relation is not the equilibrium kernel. For every fixed photon energy q, isotropic angular distributions are stationary under this Thomson operator. This displayed scalar-intensity kernel does not by itself derive the Stokes-space Thomson matrix needed for polarization.

Proof sketch. At low occupancy the commit rate between momentum cells is bilinear in the occupancies with kernel given by the squared vertex; the soft (photon energy $\ll m_e$) limit of the recovered U(1) vertex is classical Thomson scattering with the stated total cross-section and $1 + \cos^2$

angular kernel; this step is standard QED evaluated on the branch couplings and is not repeated here. The detailed-balance receipt equates the stationary weights of each reversible proposal pair. The resulting symmetric rest-frame kernel damps angular anisotropy at each fixed q. It does not mix frequencies, so its stationary kernel contains every isotropic spectrum $f(q)$ rather than a unique Bose or Planck occupancy. Momentum conservation per commit, summed over commits, gives equal and opposite momentum transfer densities to the electron–baryon fluid, with coefficient fixed by $\bar{\rho}_\gamma/\bar{\rho}_b$ bookkeeping; this is the Compton drag used in §5. □

Remark 4.2 (Polarization provenance)*.* The polarization hierarchy used below imports the standard polarized Thomson scattering matrix acting on Stokes brightness. The scalar $1+\cos^2\theta$ kernel displayed in Theorem 4.1 is its intensity projection, not a derivation of the Q/U collision terms. A source-level OPH derivation of the polarized commit kernel from the recovered vertex is not supplied here. The polarization sector and the associated 16/15 diffusion factor are textbook imports rather than consequences of the displayed operator.

Remark 4.3 (Blackbody provenance)*.* The Planck background spectrum is an initial-condition and thermal-history import in this note, not a consequence of the displayed Thomson functional. Even recoil-inclusive Comptonization conserves photon number and generically approaches a Bose–Einstein spectrum with a chemical potential. Selecting the zero-chemical-potential Planck spectrum requires the appropriate early thermalization history, including photon-number-changing channels such as double Compton scattering and bremsstrahlung, or an independently justified blackbody initial condition. An OPH derivation of the blackbody spectrum requires those channels and their rates rather than inferring spectral uniqueness from angular detailed balance.

Remark 4.4 (Provenance of the normalization)*.* $\alpha_{\rm root}$ is an exact declared-map root. Its use as the physical vertex strength is conditional on the three bridge premises in Assumption 2.2. m_e is an import (Assumption 2.2): OPH supplies no source-derived charged-lepton absolute mass, so σ_T's normalization is an explicit import. The recombination history $n_e(\tau)$ is imported under Assumption 2.4.

Remark 4.5 (OPH-specific channels)*.* Two and only two places exist where OPH kernels can modify this functional at linear order: (i) a number-changing repair-exchange channel $\Gamma_{\rm rec}(k,a)$ adds a monopole source and a spectral-distortion term; it enters $C[f]$ additively and requires an active fiber, a physical clock, and a common-parent response; (ii) a collar-packet stress $\rho_A, B_A(k,a)$ enters only the gravitational source side (the Einstein constraints in §5), never $C[f]$. With both gates closed at zero, the transport below is exactly standard.

5 The Linear Multipole Hierarchy

Theorem 5.1 (Reduction to the standard hierarchy)**.** *Linearize Theorems 3.2–4.1 on the background of Assumption 2.3 with scalar perturbations, expand the temperature brightness $F_\gamma(\mathbf{k},\hat{n},\tau)$ and polarization brightness G_γ in Legendre multipoles with $\mu=\hat{k}\cdot\hat{n}$. Writing $\delta_\gamma = F_{\gamma 0}$, $\theta_\gamma = \frac{3}{4}kF_{\gamma 1}$,*

$\sigma_\gamma = \frac{1}{2}F_{\gamma 2}$, *and* $\dot\kappa \equiv a n_e \sigma_T$, *the system is (Ma–Bertschinger 1995, eq. 63):*

$$\begin{aligned}
\dot\delta_\gamma &= -\tfrac{4}{3}\theta_\gamma + 4\dot\phi,\\
\dot\theta_\gamma &= k^2\Big(\tfrac{1}{4}\delta_\gamma - \sigma_\gamma\Big) + k^2\psi + \dot\kappa\,(\theta_b - \theta_\gamma),\\
2\dot\sigma_\gamma &= \tfrac{8}{15}\theta_\gamma - \tfrac{3}{5}kF_{\gamma 3} - \tfrac{9}{5}\dot\kappa\,\sigma_\gamma + \tfrac{1}{10}\dot\kappa\,(G_{\gamma 0} + G_{\gamma 2}),\\
\dot F_{\gamma\ell} &= \frac{k}{2\ell+1}\Big[\ell F_{\gamma(\ell-1)} - (\ell+1)F_{\gamma(\ell+1)}\Big] - \dot\kappa F_{\gamma\ell}, \qquad \ell \ge 3,\\
\dot G_{\gamma\ell} &= \frac{k}{2\ell+1}\Big[\ell G_{\gamma(\ell-1)} - (\ell+1)G_{\gamma(\ell+1)}\Big] + \dot\kappa\Big[-G_{\gamma\ell} + \tfrac{1}{2}\,(F_{\gamma 2} + G_{\gamma 0} + G_{\gamma 2})\Big(\delta_{\ell 0} + \tfrac{\delta_{\ell 2}}{5}\Big)\Big],
\end{aligned}$$

coupled to baryons,

$$\dot\delta_b = -\theta_b + 3\dot\phi, \qquad \dot\theta_b = -\frac{\dot a}{a}\theta_b + c_b^2 k^2\delta_b + k^2\psi + \frac{4\bar\rho_\gamma}{3\bar\rho_b}\,\dot\kappa\,(\theta_\gamma - \theta_b),$$

and to the Einstein constraints of Assumption 2.1,

$$k^2\phi + 3\frac{\dot a}{a}\Big(\dot\phi + \frac{\dot a}{a}\psi\Big) = -4\pi G a^2\,\delta\rho_{\rm tot}, \qquad k^2(\phi - \psi) = 12\pi G a^2(\bar\rho + \bar p)\sigma_{\rm tot},$$

where $\delta\rho_{\rm tot}$ *and* $\sigma_{\rm tot}$ *include the gated* ρ_A, B_A *contributions of Remark 4.5. These contributions are zero on the receipt-failing branch. With* $\Gamma_{\rm rec} = B_A = 0$ *this is verbatim the standard Einstein–Boltzmann system.*

Proof. For scalar intensity, insert the geodesic characteristics of Theorem 3.2 on the perturbed metric, the collision functional of Theorem 4.1, keep first order, and project on Legendre multipoles. The $G_{\gamma\ell}$ equations additionally use the imported polarized Thomson matrix stated above. The subsequent calculation is the textbook computation (Ma–Bertschinger 1995; Dodelson, *Modern Cosmology*). The two conditional OPH channels appear at their unique insertion points. □

6 Acoustic normal form, damping, and the line of sight

Proposition 6.1 (Acoustic oscillations)**.** *In the tight-coupling regime* $\dot\kappa \gg k,\ \frac{\dot a}{a}$, *define* $R \equiv 3\bar\rho_b/4\bar\rho_\gamma$ *and* $\Theta_0 = \delta_\gamma/4$. *To leading order in* $1/\dot\kappa$,

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\Big[(1+R)\,\dot\Theta_0\Big] + \frac{k^2}{3}\,\Theta_0 = -\frac{k^2}{3}(1+R)\,\psi + \frac{\mathrm{d}}{\mathrm{d}\tau}\Big[(1+R)\,\dot\phi\Big],$$

a driven oscillator with sound speed $c_s^2 = 1/[3(1+R)]$. *The WKB solution gives, at last scattering* τ_*,

$$[\Theta_0 + \psi](\tau_*) \;\propto\; \cos(k\,r_s(\tau_*)), \qquad r_s(\tau_*) = \int_0^{\tau_*} c_s\,\mathrm{d}\tau,$$

so extrema sit at $k_n\,r_s(\tau_*) = n\pi$*: compressions (odd* n*) are baryon-enhanced, rarefactions (even* n*) suppressed, and the dark-matter driving of the potentials boosts the third peak: every feature of the measured TT curve is produced by this system and by no OPH-specific term.*

Proposition 6.2 (Silk damping)**.** *At next order in* $1/\dot\kappa$ *the oscillator acquires the diffusion envelope* e^{-k^2/k_D^2} *with*

$$k_D^{-2} = \int_0^{\tau} \frac{\mathrm{d}\tau'}{6(1+R)\,\dot\kappa}\left[\frac{R^2}{1+R} + \frac{16}{15}\right],$$

the $16/15$ *arising from the polarization terms of Theorem 5.1; photon diffusion between commits is the physical mechanism, and its OPH reading is transparent: on scales crossed by a random walk of commit-free transport legs, the record field cannot maintain contrast.*

Proposition 6.3 (Line-of-sight solution). *With visibility $g(\tau) = \dot\kappa\, e^{-\kappa(\tau,\tau_0)}$, the temperature multipoles today are*

$$\Theta_\ell(k,\tau_0) = \int_0^{\tau_0} \mathrm{d}\tau \left\{ g\left[\Theta_0 + \psi + \tfrac{\Pi}{4}\right] j_\ell(k(\tau_0-\tau)) + \left(\frac{g\, v_b}{k}\right)^{\cdot} j_\ell + \frac{3}{4k^2}\,(g\Pi)^{\cdot\cdot}\, j_\ell + e^{-\kappa}\,(\dot\psi + \dot\phi)\, j_\ell \right\},$$

($\Pi = F_{\gamma 2} + G_{\gamma 0} + G_{\gamma 2}$; every unlabelled j_ℓ has argument $k(\tau_0 - \tau)$), and

$$C_\ell^{TT} = 4\pi \int \frac{\mathrm{d}k}{k}\, \Delta_\zeta^2(k) \left| \frac{\Theta_\ell(k,\tau_0)}{\zeta(k)} \right|^2 .$$

Given the conditional (A_s, n_s) of Assumption 2.5 and the imports of Assumptions 2.2–2.4, this reproduces the curve computed by the diagnostic CAMB calculation. Agreement of an independent implementation with CAMB at the sub-percent level is the numerical acceptance test.

7 Premises and Numerical Tests

Condition	Statement	Classification
Photon refinement convergence	Finite-lattice refinement Liouville: coarse-grained photon-class record transport has quotient lumpability, tightness, convergent initial laws and first conditional moments, vanishing scaled second conditional moments, generator convergence on a frozen test basis, unique geodesic characteristics, and vanishing schedule, refinement, boundary, mass-shell, and measure defects. This condition is independent of repair confluence and of the orientation-doubled label count.	simulator test below
Thomson normalization	σ_T normalization: absolute m_e is a declared import; the charged-family absolute-scale theorem is not supplied.	declared import
Recombination history	$x_e(\tau)$ recombination history from imported atomic data; contains no CMB-target information.	declared import
Numerical transfer test	Independent hierarchy implementation matches CAMB temperature and polarization spectra at stated tolerances with identical inputs.	not supplied; engineering test specified below
Tilt certificate	Conditional $n_s = 1 - P_\star/48$; source generator, weighted half, and clock normalization are not supplied.	not supplied
Amplitude + lift	A_s and radial screen-to-k map.	conditional radial theorem; finite source receipt not supplied
Kernels	Γ_{rec}, ρ_A, B_A physical receipts.	not supplied
Likelihood	Official likelihood, provenance, and data-use contract.	not supplied

Simulator test for photon refinement convergence. Freeze a smooth compactly supported test basis. At each regulator estimate the first conditional moment $\widehat{L}_r\phi$ and scaled conditional variance $\widehat{\Gamma}_r(\phi)$, together with quotient-measure, mass-shell, schedule, refinement, boundary-leakage, and collision/free-channel residuals. Pass requires $\widehat{L}_r\phi \to X_H\phi$ and $\widehat{\Gamma}_r(\phi) \to 0$ on the frozen basis with the preregistered uniform compact envelopes. Random-schedule tests are supplemental. A nonzero second-moment limit selects the diffusion alternative rather than the Liouville theorem.

8 Scope of the Conditional Reduction

The cited branch theorems supply the *scalar angular form* of the transport and locate the two conditional OPH kernels. The polarization hierarchy, blackbody background, absolute normalizations, and recombination history are imported. A physical cosmic microwave background prediction requires a source-level polarized collision bridge, blackbody and thermalization provenance, photon refinement convergence, the numerical transfer test, a physical radial lift, a finite primordial source, active-kernel evidence, and an immutable likelihood contract. This reduction certifies no initial conditions, normalizations, recombination history, or likelihood, and it reads no Planck data. Under the anti-fitting rule, the acoustic peak structure is attributable to standard plasma physics. OPH's falsifiable exposure is concentrated in the primordial inputs.

References

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- Related OPH papers: *Observers Are All You Need*; *Reality as a Consensus Protocol*; *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*; *Finite-Source Cosmology from Observer-Patch Holography.*

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